

Case Assembly Guide

Required Parts

- ☐ Waveshare ESP32-S3-Touch-LCD-1.46C board
- ☐ 3D printed case (files: [case/](#) folder or link to STL)
- ☐ USB-C cable (for power/programming)
- ☐ (Optional) 3.7V LiPo battery for portable operation

Board Documentation

- **Official Waveshare Wiki:** <https://docs.waveshare.com/ESP32-S3-Touch-LCD-1.46>
- **Schematic PDF:** Available on Waveshare wiki under "Resources"
- **Dimensions:** 52mm × 52mm (verify from schematic)

Assembly Steps

Step 1: Test Board Before Assembly

Always flash and test the board BEFORE mounting in the case:

1. Upload FlightRadar.ino
2. Verify display shows "FlightRadar" splash
3. Test touch responsiveness
4. Connect to WiFi

Step 2: Case Orientation

- **Display faces up** (round LCD visible)
- **USB-C port** should align with case opening
- **Boot button** should be accessible (for recovery)

Step 3: Mounting

- Use M2.5 screws (4 locations)
- Do not overtighten - plastic threads strip easily
- Ensure no pressure on LCD flex cable

Step 4: Cable Management

- Route USB-C cable through strain relief
- Battery (if used) fits in compartment behind board

Troubleshooting Assembly Issues

Issue	Solution
Display doesn't fit	Check case orientation - board only fits one way
Can't access BOOT button	Use paperclip through case hole, or disassemble
Case blocks WiFi	Ensure antenna area (top of board) is not enclosed in metal

3D Print Settings (if applicable)

- Layer height: 0.2mm
- Infill: 20%
- Supports: Yes (for USB-C opening)
- Material: PLA or PETG